Online Resource 1: Proposed Empirical Designs

Supplement to *Grounding Without Corrective Control: Truth-Tracking Profiles for Large Language Models*

Journal: *Synthese*

1 Purpose

This resource specifies three component tests proposed in the accompanying article: a crossed route-by-task design, a comparison that varies checking-route independence, and a temporal test of live answerability. The designs don't jointly distinguish the decomposition from an unrestricted effective-coverage description. Their purpose is to test whether a decomposition fixed before the outcomes are observed supports selective transfer, diagnosis, and intervention. No empirical results are reported; the numerical results below come only from constructed data.

Table 1 states the inferential role of each design. It treats the tests as validation arguments: an observed contrast supports a specified component only under the declared architectural boundary, auxiliary assumptions, and manipulation checks. None is a crucial experiment against every informational redescription.

2 Fake-data stress tests of the decomposition

Constructed data can audit a proposed model comparison before it is used on real systems (Gelman et al., 2020). Their causal structure is chosen, so they cannot show that the decomposition is true. The exercise asks whether the analysis can recover a known architectural advantage, whether a simpler prespecified model can beat it, and whether predictive gains improve intervention choice.

Let p_0 be an arrangement's probability of an initially correct answer. For a correction opportunity, let x, d, and i represent target access, detectability, and checking-route independence as operationalized by task-specific conditional non-overlap, and let $c = xdi$ be the route's conditional signal. Let a and u represent attribution and uptake, f the false-signal probability on an initially correct answer, ρ the probability that an exercised correction succeeds, and η the probability that an unwarranted intervention causes harm.

The structural data-generating model is

$$\begin{aligned} P(\text{repair} \mid \text{initial error}) &= cau\rho, \\ P(\text{harm} \mid \text{initially correct}) &= fau\eta, \\ P(\text{final correctness}) &= p_0(1 - P(\text{harm})) + (1 - p_0)P(\text{repair}). \end{aligned}$$

Test	Decomposition	Strongest coverage reply	Evidential role
Route by task	A supplied route should help most where the task requires it.	Conditional or effective coverage predicts the same pattern by distinguishing kinds of task-relevant information.	Tests the proposed route carving and selective transfer, not uniqueness.
Checker relation	Lower conditional error overlap should improve repair when marginal accuracy and baseline inputs are matched.	Conditional coverage predicts the advantage by representing information added given generator error.	Tests checking-route independence; distinguishes only a fixed input-coverage comparator.
Temporal update	A live route should transmit a source update that a frozen record can't.	Dynamic coverage predicts the same result because follow-up information differs.	Operationalizes live answerability; doesn't discriminate against dynamic coverage.

Table 1: Predictions and limits of the three component tests.

This product assigns a probability to success along one deliberately simple serial path. Every stage has to succeed on a correction opportunity, so strength at one stage can't compensate for a missing stage. The specification neither ranks arrangements globally nor assumes that the feature measures are statistically independent.

The fixed conditional-coverage comparator omits a and u, using $c\rho$ and $f\eta$; the input-coverage comparator uses target access alone for repair. Each model estimates one repair and one harm multiplier from training cells. To isolate the route comparison, every model receives the same true p_0. This removes baseline-estimation error and makes the exercise easier than an empirical application, which would have to estimate p_0 with uncertainty.

These specifications are test vehicles. A stronger comparator specified before the outcomes are observed should replace them. If it predicts or guides intervention as well with fewer commitments, the decomposition loses its projective claim. Adding a and u to the coverage model would reproduce World A, but would make it an alternative parameterization of the same serial architecture rather than an independent explanation.

The simulation draws 30 heterogeneous arrangements, record and measurement tasks, and five equal-cost choices: no intervention, retrieval, measurement, independent checking, and enforced uptake. Each arrangement–task–intervention cell contains 120 trials. Four intervention–task combinations are withheld from fitting. The models are compared on those cells by logarithmic loss for repair conditional on initial error and by regret from choosing an intervention other than the truly best one. Figure S1 shows paired differences over 200 replications from a fixed master seed. Negative repair-loss differences favour the decomposition; positive optimal-choice differences do.

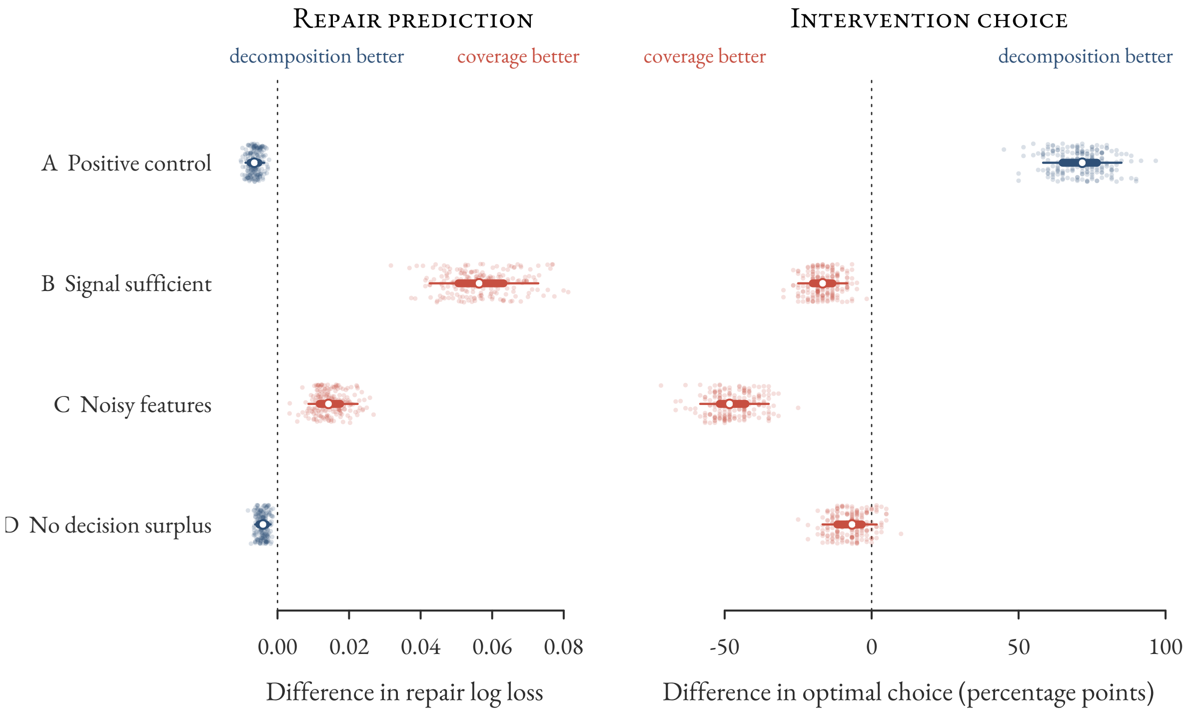

Figure S1: Paired differences between the decomposition and conditional coverage across 200 constructed replications. Points show replications; thin and thick lines span the central 90 and 50 percent, and circles mark medians. Blue favours the decomposition and coral favours conditional coverage, redundantly marked by position relative to zero. Each world is a separate stress condition: A is a positive control, while B–D specify conditions under which the decomposition can lose.

Figure S1 reports the audit. World A is a positive control: attribution and uptake were built into the generating process and omitted from the comparator. Worlds B–D carry the substantive burden. Conditional coverage is sufficient in B. In C, route structure generates the data, but noisy feature measurements make the decomposition choose the optimal intervention 47.23 percentage points less often. In D, it predicts repair better without improving intervention choice. The decomposition earns its extra complexity only when its distinctive features are causally important, measurable, and useful for prediction or decisions.

Differences in final-accuracy loss are much smaller because initially correct answers dominate that aggregate, which is why the article treats repair conditional on error and harmful intervention as the more diagnostic quantities. The repair and regret comparisons in each of the first three worlds have the same direction in all 200 replications; in the fourth, decision regret is effectively tied. The reproducible R script and machine-readable results accompany this resource.

3 A crossed route-by-task design

The worked example uses clinical questions. One item set asks for recommendations recoverable from guidelines or institutional records. A paired set uses the same disease area and answer format and requires a patient-specific lab value, observed sign, dose calculation, or intervention-sensitive causal judgment. The prediction is a retrieval gain on the record-recoverable set and little transfer to the patient-specific set unless the needed measurement or tool route is supplied. Retrieval producing the same gain on both sets would count against the proposed distinction between routes. The design parallels validity argumentation in measurement: inferences from performance are licensed relative to a purpose and tested by varying method while holding the target construct fixed (Kane, 2013; Messick, 1989).

Matching the sets on difficulty cues isn't sufficient. An intervention that supplies a missing lab value also makes the item easier, so a difficulty explanation and a route explanation both predict a gain. Equating measured baseline accuracy weakens the simplest difficulty account without eliminating differences in discrimination, ceiling structure, response variance, latent subskills, or treatment-effect heterogeneity. Matching on noisy estimates also invites regression to the mean. Use baseline performance as a covariate and design diagnostic, not as evidence that difficulty has been removed.

The primary design is crossed rather than merely matched. Build item families in which the relevant route can be supplied or withheld within closely related items, then cross route availability against task requirement. Estimate the interaction in a hierarchical model with varying effects for items, domains, prompt templates, and systems. Declare in advance the smallest interaction that would matter, report its interval, and run reasonable alternatives for item matching, scoring, and exclusion rather than relying on one sample.

In the single-pair implementation, the comparison uses two named chat arrangements that differ only in whether retrieval over a fixed corpus is available, with the prompt template held constant. The estimate is conditional on that pair; variation across model families and deployment setups lies outside the estimand and belongs to a multi-system extension. The population is a clinical item set split into record-recoverable and patient-specific halves. Raters classify the items from their text without seeing system output, and the classification is frozen before the first run.

The primary outcome is substantive correctness of the recommendation, scored blind, which applies to both halves of the item set. Whether verbatim anchors resolve is secondary and applies only within the record-dependent half. Making anchor resolution primary would manufacture the interaction because patient-specific items have no corpus anchors to resolve.

The worked prediction is that the retrieval gain on record-recoverable items exceeds the gain on patient-specific items by at least ten percentage points. Ten is illustrative, not derived. A defensible threshold needs a decision basis: the cost of a wrong recommendation, the cost of retrieval latency, and what a deployment would do differently at nine points rather than eleven. Fix the value before outcomes are observed so that the prediction can fail.

The interval is interpreted against that threshold rather than zero. Concentration above ten points supports the worked claim; concentration near zero or on the wrong side counts against it. An interval extending from below zero to well above ten describes an uninformative experiment, not evidence either way. The threshold isn't revised after the effect becomes visible.

A stronger implementation crosses both interventions with both task types. Retrieval should selectively help record-dependent items; a measurement tool should selectively help items requiring measurement. This double dissociation tests the proposed route carving, though a relevance-sensitive information account can predict the same qualitative interaction.

4 Checking-route independence

Compare two checkers: one from the generator's lineage and the other trained and calibrated on separate data and feedback. Give them the same candidate output and task evidence supplied from outside the checkers. Match standalone accuracy, confidence, latency, response format, and revision policy as closely as possible. Measure detection and correction conditional on generator error, especially under perturbations expected to produce common-mode failure.

Verify rather than assume the match. Use held-out calibration items to compare standalone accuracy, confidence calibration, and latency, and route both checker outputs through the same response format and revision policy. The primary cross-route quantity is conditional error overlap: how often the checker repeats the generator's error when the generator is wrong. Report it together with false interventions on generator outputs that were already correct.

For this comparison, input coverage is measured before either checker produces a task-specific signal. Checker outputs are downstream consequences of the architecture, not part of baseline input coverage. If coverage is measured after a checker responds, the predicted advantage is already a conditional-coverage effect.

The decomposition predicts an advantage for the checker with lower conditional error overlap. Calibration history matters only insofar as it predicts that overlap; if admissible perturbations produce no difference, the decomposition has no reason to privilege one history. An input-coverage account predicts no advantage once externally supplied task information and standalone accuracy are matched. A conditional-coverage account can predict one by representing the joint error structure, but that prediction incorporates the relation that checking-route independence names.

5 A temporal test of live answerability

Give two arrangements the same target information at the outset, one holding a frozen pasted record and the other holding a live route to its source. In the update arm, correct the source record and ask a follow-up whose answer turns on the correction. In the control arm, leave the record unchanged. The live route should track the update in the first arm and match the frozen copy in the second. Figure S2 summarizes the design.

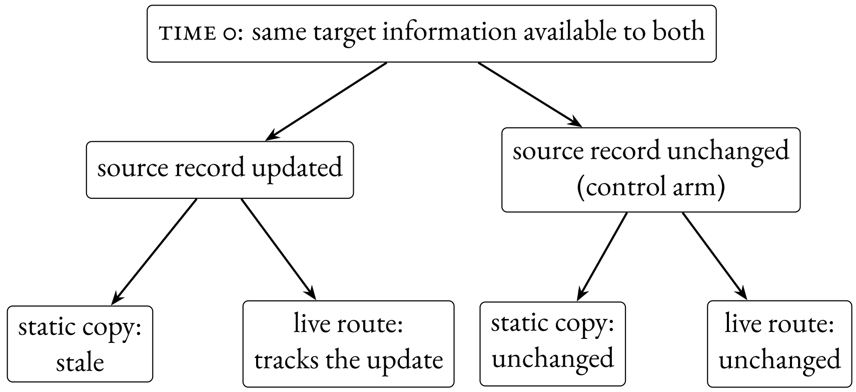

Figure S2: A temporal test of live answerability. At time 0, both arrangements have the same target information. In the update arm, the source record is corrected: the static copy stays stale while the live route transmits the correction. In the control arm, neither should change. No outcome magnitudes are implied.

A dynamic coverage account predicts the same pattern because the arrangements receive different information at follow-up. The manipulation operationalizes live answerability and isolates updateability from initial content equivalence; it doesn't adjudicate between the decomposition and dynamic coverage.

6 Manipulation checks and failure conditions

Failure of a declared route-specific interaction or update effect counts against the decomposition only after confirming that the focal route was invoked and that its output reached production in time to affect the answer. A stale index, failed tool call, or ignored checker signal shows that the intended manipulation wasn't instantiated. Once those checks pass, failure of the declared contrast is evidence that the proposed route carving or corrective feature has been misdescribed.

References

Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.-C., & Modrák, M. (2020). *Bayesian workflow*. https://doi.org/10.48550/arXiv.2011.01808

Kane, M. T. (2013). Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, *50*(1), 1–73. https://doi.org/10.1111/jedm.12000

Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–103). American Council on Education; Macmillan Publishing Company.